

PRODUCT REQUIREMENTS DOCUMENT



AI-Powered Home Construction Management Platform

B	Blueprint	<i>AI Design</i>
U	Understanding	<i>Insights</i>
I	Intelligence	<i>AI Engine</i>
L	Lifecycle	<i>Tracking</i>
D	Design	<i>Interior & Finish</i>

A tech-enabled residential construction platform for the Indian market — combining AI-generated, Vastu-aware floor plans with real-time site monitoring, escrow-based finance, predictive scheduling, and integrated interior design.

Version	1.0
Date	25 May 2026
Status	Draft for Engineering Review

Author	Product Team
Audience	Engineering, Architecture, Design
Project Code	HOMEBUILD-V1

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1. Executive Summary

HomeBuild is a tech-enabled residential construction management platform purpose-built for the Indian homeowner. It serves first-time builders, plot owners, and homeowners across Tier 1, Tier 2, and Tier 3 cities with affordable to mid-range residential construction — bringing the predictability, transparency, and quality of a tech-led platform to a market historically dependent on local contractors and word-of-mouth referrals.

The name HomeBuild — and specifically the BUILD acronym — captures the five pillars of the product:

- **B — Blueprint · AI Design:** Generative floor plans tailored to plot dimensions, budget, and Vastu Shastra principles
- **U — Understanding · Insights:** Data-driven project insights including budget consumption, schedule variance, and quality trends
- **I — Intelligence · AI Engine:** Computer vision quality monitoring, forecasting, and Vastu compliance scoring engine
- **L — Lifecycle · Tracking:** Real-time site progress tracking, milestone-based escrow payments, and Gantt-based scheduling
- **D — Design · Interior & Finish:** Post-structural interior design tools, finish material catalogues, and styling recommendations

The platform organises every project around a circular four-module compass interface — Design, Construction, Finance, and Schedule — corresponding to the four directional pointers (North, East, South, West). Each module is purpose-built for a specific job-to-be-done in the residential construction lifecycle, from initial floor plan generation through to final handover and interior finishing.

Key Product Differentiators

- AI-powered Vastu-compliant floor plan generation tailored to Indian construction standards
- Computer-vision-based site quality monitoring through smartphone photos (no expensive IoT)
- Regional-language-first interface for accessibility across India
- Material procurement integration with platform-negotiated rates
- Annual maintenance contract (AMC) as a post-construction revenue stream

Strategic Positioning

HomeBuild is built for the Indian homeowner who wants a transparent, AI-assisted construction experience. The platform's strength lies in Vastu compliance, regional language support, predictable milestone-based pricing, and integrated interior design — making it equally valuable in metros and Tier 2/3 cities where local trust and cultural alignment matter most.

2. Product Vision & Goals

2.1 Vision Statement

Empower every Indian homeowner — from a first-time builder in cities to a seasoned plot-owner in metros — to construct their dream home with the predictability, transparency, and quality of a tech-led platform, at a price they can afford.

2.2 Product Goals (12-Month Horizon)

Table: Year 1 Product Goals and Measurable Targets

undefined	undefined	undefined	undefined
1	Validate product-market fit in pilot city	Completed projects	50 in 6 months
2	Demonstrate AI design value	% of customers using AI plans	70%+
3	Achieve operational reliability	On-time milestone completion	85%+
4	Build a defensible contractor network	Active vetted contractors	200+ across 3 cities
5	Reach unit economic viability	Margin per project	₹3.5L average
6	Drive organic growth	% projects from referral	30%+

2.3 Non-Goals (Explicitly Out of Scope for v1)

- Commercial construction (offices, warehouses, retail)
- High-rise residential (G+4 and above)
- Land acquisition or plot brokerage
- Home loan origination (referral-only)

3. Market Context

3.1 Market Opportunity

The Indian residential construction market is estimated at over ₹6 lakh crore annually, with the individual home construction (IHC) segment alone representing ₹1.5–2 lakh crore. Tech penetration in this segment is below 5%, leaving substantial whitespace for platform-based players that can deliver predictability, transparency, and quality at scale. Tier 2 and Tier 3 cities — where construction volumes are growing fastest — are particularly underserved by tech-led solutions.

3.2 Why Now

Table: Market Trends Justifying Product Timing

undefined	undefined
Rising disposable income	More homeowners can afford to commission full-home construction rather than buying ready apartments
Smartphone & UPI penetration	Customers across all tier cities are comfortable transacting digitally and tracking progress online
Generative AI maturity	LLMs can now produce realistic, Vastu-aware floor plans in seconds — previously impossible at this price point
Computer vision affordability	Quality monitoring via phone-camera photos is now production-grade without expensive on-site IoT
Escrow infrastructure	RBI-regulated escrow services (RazorpayX, Cashfree) enable trust-based milestone payments
Regional language LLMs	Indic language support has reached production quality, unlocking Tier 2/3 markets

3.3 Target Customer Segments

- **Primary:** Middle-income homeowners across Tier 1, 2, and 3 cities
- **Secondary:** Plot owners building investment properties (single G+1 or G+2 residences for rental income)
- **Tertiary:** NRI customers building family homes in India who need remote monitoring and trusted execution

4. User Personas

Rajesh Sharma — Primary Customer

Homeowner / Plot Owner · Age 38 · Coimbatore

Software engineer turned local entrepreneur. Owns a 2,400 sq.ft inherited plot in a developing neighbourhood. Wants to build a G+2 home for his family with shop space on the ground floor.

Goals

- Build a quality home within ₹45L budget
- Ensure Vastu compliance (family priority)
- Avoid the typical contractor headaches (delays, cost overruns)
- Track progress remotely while at work

Pains

- Doesn't trust local contractors with full payment upfront
- Has never managed a construction project
- Worried about quality issues he won't be able to spot
- Family pressure for monthly progress updates

Suresh Patel — Project Manager (Internal)

HomeBuild Project Manager · Age 32 · Multi-site (3–5 projects)

Civil engineering background, 8 years experience. Manages multiple projects simultaneously from his phone and laptop. Acts as the bridge between customer, contractor, and platform.

Goals

- Reduce site visits through remote monitoring tools
- Catch quality issues before they become rework
- Keep customers informed without manual reporting
- Hit milestone deadlines for escrow releases

Pains

- Spends 40% of time on status updates that could be automated
- Misses subtle quality issues in photos sent by site supervisors
- Manually reconciling contractor invoices with milestones

Mohammed Ali — Site Supervisor

Field Operations (Contractor Side) · Age 29 · On-site daily

Diploma in civil engineering. Manages 8–15 workers daily on a single site. Uses a basic Android phone and is more comfortable in Hindi/regional language than English.

Goals

- Upload daily progress quickly (under 2 minutes)
- Get clear instructions from the project manager
- Receive timely payments without chasing

Pains

- Complicated apps slow him down
- Poor internet connectivity at most sites
- Doesn't always know what photos are 'good enough'

5. Functional Requirements

The product is organised into a central dashboard (the 'Compass') with four primary modules. The compass UI presents these as four cardinal-direction nodes: North (Design), East (Construction), South (Finance), West (Schedule).

Compass Home Screen

Upon login, the customer sees a project header (project name, location, plot size, configuration), a status badge (active/paused/completed), and four large circular buttons arranged in cardinal positions around a central HomeBuild branded orb. Tapping any node enters that module. The home screen also displays summary KPIs (overall completion %, days remaining, current phase).



Figure: Home dashboard — the compass interface with four cardinal modules. The center orb shows the HomeBuild brand mark, while the four directional nodes represent Design (N), Construction (E), Finance (S), and Schedule (W). Status cards at the bottom show project health at a glance.

5.1 Design Module — AI Floor Plan Generation

Module Identifier: NORTH | Color: Blue (#3B82F6) | Owner: AI/ML team

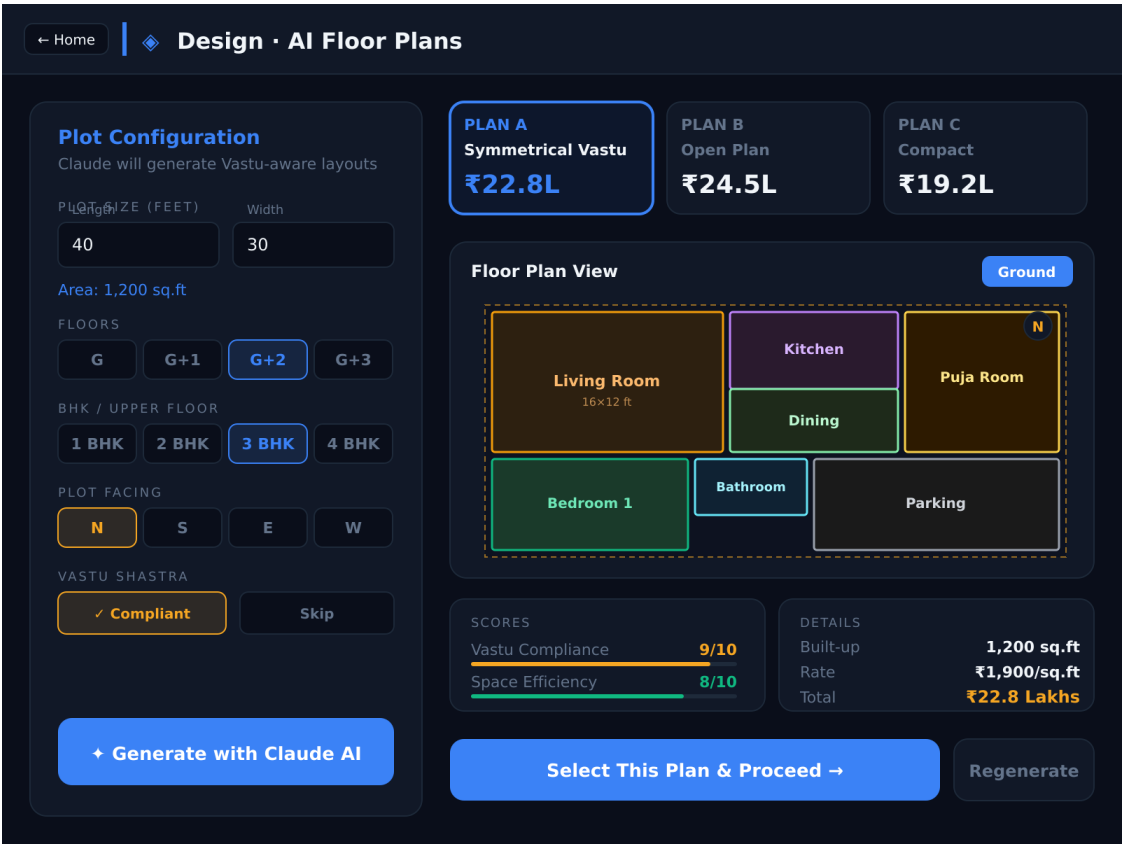


Figure: Design module — plot configuration on the left, AI-generated floor plan options on the right. Each plan shows cost, Vastu and efficiency scores, and a colour-coded visual floor plan with room dimensions.

5.1.1 Purpose

Allow homeowners to input their plot parameters and instantly receive 3 distinct AI-generated floor plan options, each with Vastu compliance scoring, cost estimates, and room-by-room dimensions, before they commit to a construction project.

5.1.2 User Stories

- **US-D-01** As a homeowner, I want to enter my plot dimensions and preferences, so I can see multiple floor plan options without hiring an architect upfront.
- **US-D-02** As a Vastu-conscious customer, I want each plan to show a Vastu compliance score, so I can choose a layout aligned with my beliefs.
- **US-D-03** As a budget-conscious user, I want to see total cost and cost-per-sq.ft for each plan, so I can choose what fits my finances.
- **US-D-04** As a multi-floor builder, I want to view each floor's layout separately, so I can verify how the home stacks vertically.

5.1.3 Input Parameters

Table: Design Module — Input Parameters & Validation Rules

undefined	undefined	undefined	undefined
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Plot Length (ft)	Integer	Yes	Range: 15–200
Plot Width (ft)	Integer	Yes	Range: 15–200
Number of Floors	Enum	Yes	G, G+1, G+2, G+3
BHK per Upper Floor	Enum	Yes	1, 2, 3, 4
Parking Spots	Enum	Yes	0, 1, 2, 3
Plot Facing	Enum	Yes	N, S, E, W
Architectural Style	Enum	Yes	Modern, Traditional, Contemporary, Minimal
Budget Range	Enum	Yes	Economy, Mid, Premium, Luxury
Vastu Compliance	Boolean	Yes	Toggles strict Vastu rules
Extra Spaces	Multi-select	No	Balcony, Puja Room, Study, Utility

5.1.4 AI Generation Logic

The system passes the input parameters to a prompted LLM call (Claude Sonnet or equivalent) that returns three structured floor plan variants. Each variant follows a different design philosophy:

- **Plan A — Symmetrical Vastu:** Maximum Vastu compliance, central courtyard, equal sunlight on all sides
- **Plan B — Open Plan Contemporary:** Modern open kitchen-living, large windows, premium aesthetic
- **Plan C — Compact Efficient:** Minimal dead space, budget-friendly, maximum usable area

5.1.5 AI Output Structure

Each AI generation call returns a structured response containing an array of three plans. Every plan includes the following fields:

Table: Floor Plan Response Fields

undefined	undefined	undefined
Plan ID	Letter	Unique identifier — A, B, or C
Plan Name	Text	Descriptive plan name (e.g. 'Symmetrical Vastu')
Vastu Score	Number 1–10	How well the layout follows Vastu Shastra principles
Efficiency Score	Number 1–10	How well usable space is maximised within the plot
Total Built-up Area	Square feet	Sum of all enclosed room areas across all floors
Cost per Square Foot	Indian Rupees	Rate band based on selected budget tier
Total Cost	Indian Lakhs	Total construction cost (built-up area × rate)

Highlights	Text list	Three key features describing the plan's strengths
Vastu Notes	Text	Brief explanation of Vastu zoning applied
Floors	List	One entry per floor (Ground Floor, First Floor, etc.)
Rooms (per floor)	List	Each room has a name, position, width, and height in feet

Room positions use a simple grid coordinate system measured in feet from the top-left corner of the plot. Each room has an X position (distance from the left edge), Y position (distance from the top edge), width, and height. This data is then converted into a visual SVG floor plan for display to the customer.

5.1.6 Vastu Compliance Engine

The AI prompt enforces standard Vastu Shastra room placement rules. The engine must respect:

- Kitchen → South-East zone (Agni)
- Master Bedroom → South-West zone (Earth element)
- Puja Room → North-East zone (Ishanya)
- Living Room → North or North-East zone
- Toilets → North-West or West zone
- Staircase → South or South-West zone
- Main door → North, East, or North-East preferred

5.1.7 Floor Plan Rendering

Each generated plan is rendered as an SVG floor plan with the following visual conventions:

- Rooms colour-coded by type (bedrooms blue, kitchen purple, living orange, etc.)
- Room labels with dimensions overlaid (e.g., '14×12 ft')
- Plot boundary shown as dashed orange line with overall dimensions
- North arrow indicator in top-right corner
- Grid background at 1-foot resolution
- Multi-floor plans accessible via tab selector (Ground / First / Second floor)

5.1.8 Acceptance Criteria

- Three distinct plans returned within 8 seconds of submission
- Every room fits within plot boundary (no room with $x+w > \text{plot length}$, $y+h > \text{plot width}$)
- No two rooms overlap (room intersection check)
- Total built-up area is between 60–80% of plot × floors (FAR compliance)
- If Vastu is enabled, Vastu score must be $\geq 7/10$ for all 3 plans
- Cost calculation uses correct ₹/sq.ft band based on budget input (Economy: ₹1,400–1,700; Mid: ₹1,800–2,100; Premium: ₹2,200–2,600; Luxury: ₹2,700+)

5.2 Construction Module — AI Site Monitoring

Module Identifier: EAST | Color: Green (#10B981) | Owner: Computer Vision team

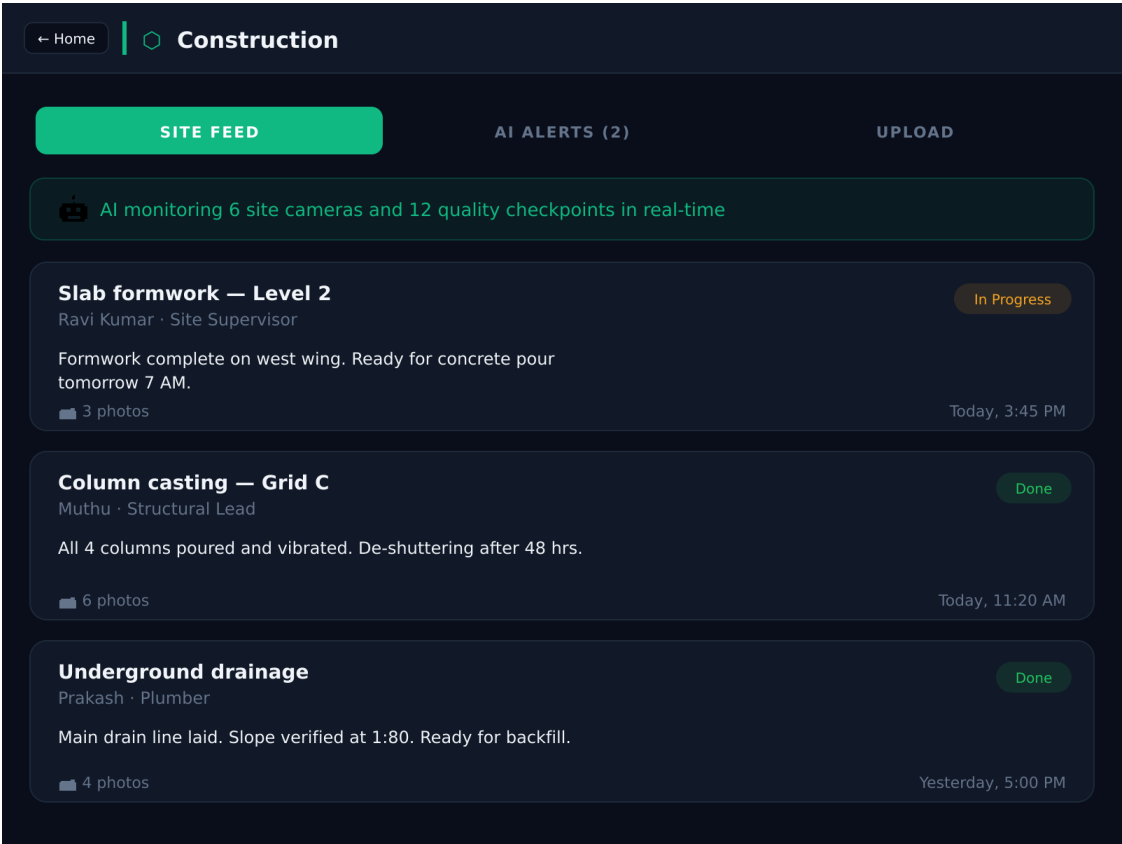


Figure: Construction module — daily site updates from supervisors with progress photos and AI-driven quality alerts. Each update shows the work category, status badge, supervisor notes, and attached photo count.

5.2.1 Purpose

Provide a real-time view into on-site progress through site supervisor photo uploads, paired with an AI-based quality monitoring layer that automatically flags potential issues to the project manager.

5.2.2 Sub-Components

- **Site Feed:** Chronological stream of daily updates from site supervisors with photos, notes, and task tags
- **AI Alerts:** Real-time notifications of quality issues detected by computer vision analysis of uploaded photos
- **Photo Upload:** Site supervisor interface for uploading progress photos with structured metadata

5.2.3 Computer Vision Quality Checks

The CV model is trained to detect deviations across the following construction quality dimensions. Each check returns a confidence score; alerts trigger above a configurable threshold (default 0.75).

Table: Computer Vision Quality Check Categories

undefined	undefined	undefined
Reinforcement Spacing	Rebar spacing irregularities in columns and slabs	High
Concrete Curing	Visible cracks, dry spots, inadequate moisture coverage	High
Formwork Alignment	Tilted or out-of-plumb shuttering before pour	High
Brickwork Bonding	Improper bond pattern, missing mortar joints	Medium
Plaster Quality	Uneven thickness, hollow spots, surface cracks	Medium
Safety Compliance	Workers without helmets, missing scaffolding, unsafe ladders	High
Material Storage	Cement bags exposed to rain, steel rusting, sand contamination	Low
Site Cleanliness	Debris piles, blocked access, scattered tools	Low

5.2.4 Alert Workflow

- CV model processes uploaded photo within 30 seconds
- If issue confidence > 0.75, an alert is generated with severity, location pin, and description
- Project Manager receives push notification + in-app alert
- PM reviews alert and either dismisses (false positive), assigns to contractor for fix, or escalates to client
- Resolved alerts archived with audit trail (resolution photo, signed-off by)

5.2.5 Photo Upload Specifications

- Accepts JPG, PNG, HEIC up to 10MB per photo
- Auto-compression to 1080p before upload (to handle low-bandwidth sites)
- Required metadata: work category, location on site (room/area), brief notes
- Optional metadata: workers present, materials used, milestone reference
- Offline capture supported — uploads when connectivity restored

5.2.6 Acceptance Criteria

- Site feed updates appear in customer view within 60 seconds of upload
- CV analysis returns within 30 seconds of photo upload
- False positive rate of alerts must be ≤15% (measured weekly)
- Critical alerts (High severity) reach PM within 2 minutes via push notification

5.3 Finance Module — Escrow & Budget

Module Identifier: SOUTH | Color: Amber (#F59E0B) | Owner: Payments / FinTech team

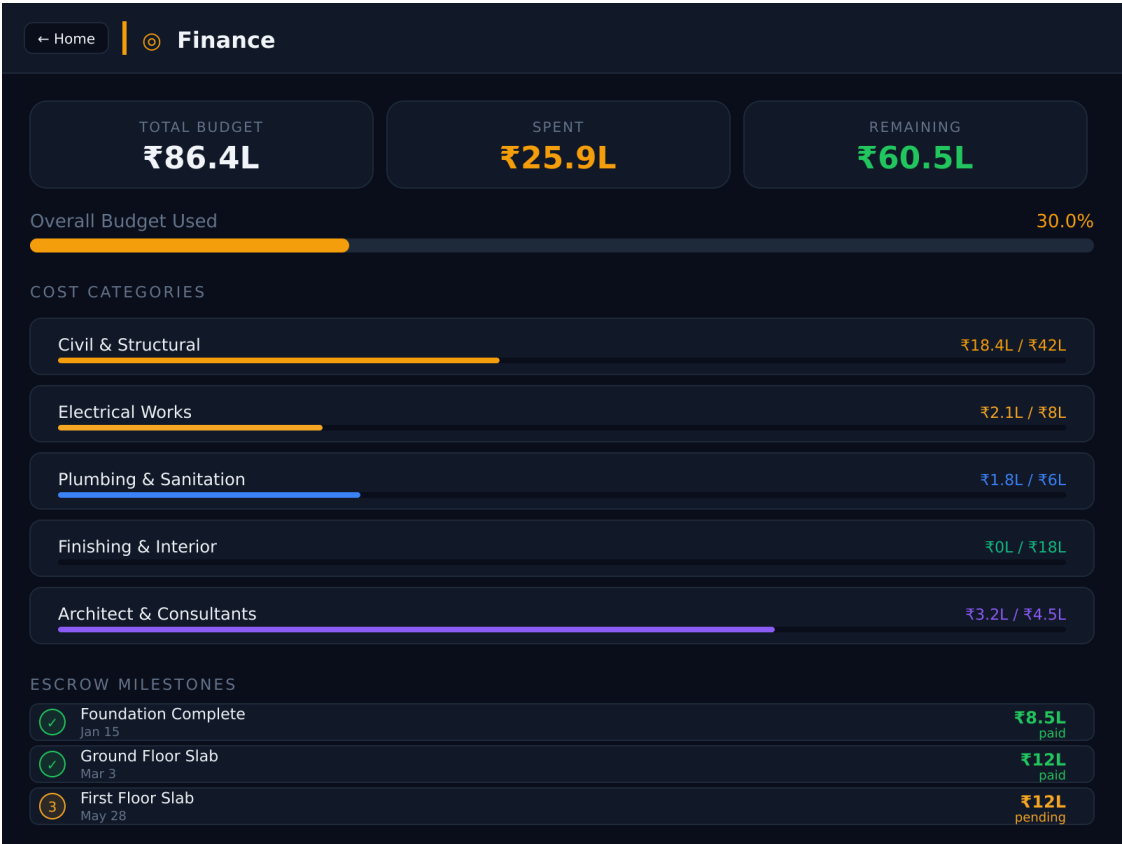


Figure: Finance module — total budget vs spent breakdown across cost categories, with milestone-based escrow payment tracking. Each milestone has a verified release status and corresponding payment amount.

5.3.1 Purpose

Provide end-to-end transparency on project finances — including category-level budget tracking, milestone-based escrow payments, and audit-ready spending records — for both the customer and the project team.

5.3.2 Cost Categories (Default Template)

Table: Default Cost Category Allocation Template

undefined	undefined	undefined
Civil & Structural	45–50%	Concrete, steel, bricks, sand, labour
Electrical Works	8–10%	Wiring, fixtures, panels, switches
Plumbing & Sanitation	6–8%	Pipes, fittings, sanitary ware
Finishing & Interior	18–22%	Tiles, paint, doors, windows, flooring
Architect & Consultants	4–6%	Design fees, structural consultant, supervision

Contingency	10%	Reserve buffer for overruns
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5.3.3 Escrow Milestone Structure

Customer funds are held in an escrow account and released only upon verified milestone completion. Each milestone has a predefined % of total project cost and verification criteria.

Table: Escrow Milestone Release Structure

undefined	undefined	undefined
Foundation Complete	18%	Sub-structure cast, PM site inspection passed
Ground Floor Slab Cast	22%	Concrete poured and cured 21 days, structural sign-off
First Floor Slab Cast	20%	Upper slab cast, all columns verified vertical
Roof Slab Cast	15%	Final slab cast, water tank base ready
Brick & Plaster Complete	12%	All walls plastered, openings finalized
Final Handover	13%	Painting, flooring, MEP done, snag list cleared

5.3.4 Payment Provider Integration

- Primary integration: RazorpayX Escrow / Cashfree Escrow
- Customer onboarding: KYC + bank verification
- Release triggers: PM digital sign-off + customer dual-approval for releases >₹5L
- Auto-reconciliation against milestone schedule

5.3.5 Acceptance Criteria

- Budget vs spent visible at category level with progress bars
- Each milestone shows status: paid / pending / upcoming with ₹ amount and date
- Escrow transactions are logged immutably with PDF receipt generation
- Overall budget consumption % updated in real-time

5.4 Schedule Module — Gantt & Delay Forecast

Module Identifier: WEST | Color: Violet (#8B5CF6) | Owner: Project Operations team

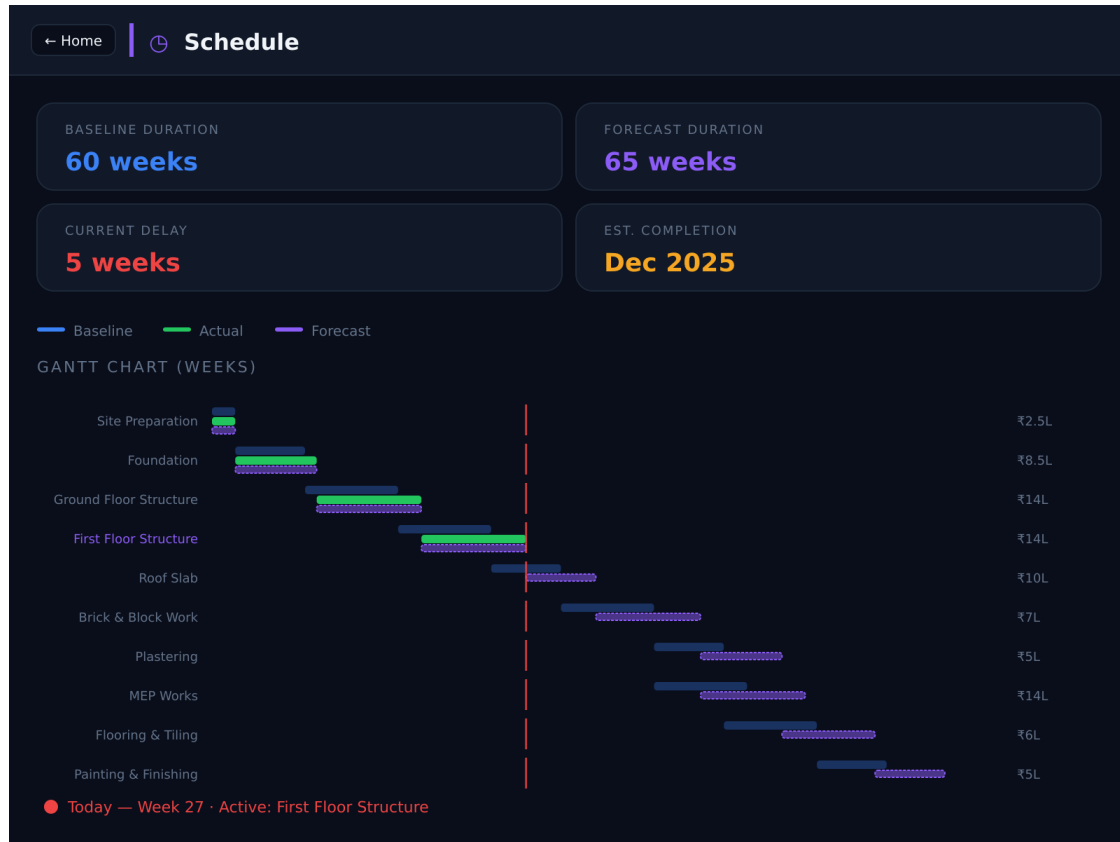


Figure: Schedule module — Gantt chart showing baseline (blue), actual (green), and forecast (violet) timelines overlaid for each construction phase. The red vertical line marks the current week, with KPI cards showing total project duration and current delay.

5.4.1 Purpose

Visualise project schedule with three overlaid timelines — baseline (originally planned), actual (executed), and forecast (predicted future) — along with delay root-cause analysis and cost impact projections.

5.4.2 Phase Structure (Default 60-week Template)

Table: Default 60-Week Construction Phase Template

Phase	Baseline Duration	Cost
Site Preparation	Week 1–2	₹2.5L
Foundation	Week 2–8	₹8.5L
Ground Floor Structure	Week 8–16	₹14L
First Floor Structure	Week 16–24	₹14L
Roof Slab	Week 24–30	₹10L
Brick & Block Work	Week 30–38	₹7L

Phase	Baseline Duration	Cost
Plastering	Week 38–44	₹5L
MEP Works	Week 38–46	₹14L
Flooring & Tiling	Week 44–52	₹6L
Painting & Finishing	Week 52–58	₹5L
Final Handover	Week 58–60	₹1L

5.4.3 Forecast Algorithm

The system uses a rule-based forecasting engine that adjusts future phase durations based on:

- Current cumulative delay (actual vs baseline at current week)
- Seasonal factors (monsoon weeks 22–36 add 15% buffer)
- Pending alerts that may cause rework (each high-severity alert adds 3–7 days)
- Resource bottlenecks reported by PM (e.g., RMC delivery delays)

5.4.4 Delay Analysis

For each completed or active phase that has overrun its baseline, the system surfaces:

- Root cause (Monsoon, material delay, design change, contractor capacity)
- Weeks added
- Additional cost incurred (in ₹L)
- Whether the delay is recoverable in subsequent phases

5.4.5 Acceptance Criteria

- Gantt chart shows all three timeline overlays (baseline blue, actual green, forecast violet)
- Today marker (red vertical line) updates daily
- Each phase shows individual cost
- Delay analysis updated whenever a phase finishes or new alert raised
- Forecast accuracy: predicted completion within ± 2 weeks of actual (measured at 75% project completion)

6. Non-Functional Requirements

Table: Non-Functional Requirements Summary

undefined	undefined
Performance	Page load < 2s on 4G; AI floor plan generation < 8s; photo upload < 5s on 3G with auto-retry
Availability	99.5% uptime SLA; planned maintenance windows announced 48 hours in advance
Scalability	Architecture must support 10,000 concurrent users and 50,000 active projects within 24 months
Security	Encryption at rest (AES-256) and in transit (TLS 1.3); SOC 2 Type II compliance roadmap
Localisation	Multi-language UI: English, Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi, Bengali, Gujarati, Punjabi, Oriya
Accessibility	WCAG 2.1 AA compliance for web; native mobile accessibility services support
Mobile-first	Site supervisor interface optimised for Android 8+ on low-end devices (2GB RAM)
Offline mode	Critical site supervisor functions (photo capture, notes) work offline with sync on reconnect
Data residency	All Indian customer data stored in Indian data centres (Mumbai, Hyderabad regions)
Audit Logs	All financial transactions and milestone sign-offs immutably logged for 7 years

7. Technical Architecture

7.1 High-Level System Architecture

HomeBuild follows a layered microservices architecture with three distinct tiers — the Client Layer (mobile and web applications), the Service Layer (modular backend services), and the Data Layer (persistent storage and caching). The architecture is designed to scale horizontally, handle intermittent connectivity for site supervisors, and integrate cleanly with third-party AI, payment, and notification providers.

Client Layer

Three distinct client applications serve the three user personas. Customers use a React Native mobile app to track their projects, view AI floor plans, and approve milestone payments. Project Managers use a Next.js web dashboard to oversee multiple projects, review alerts, and coordinate with contractors. Site Supervisors use a lightweight Android-only application optimised for low-end devices and intermittent internet connectivity.

API Gateway Layer

All client traffic flows through a central API gateway that handles authentication, rate limiting, request routing, and SSL termination. The gateway also enforces tenant isolation and routes requests to the appropriate backend service based on the API endpoint.

Service Layer

Each of the four modules (Design, Construction, Finance, Schedule) has its own dedicated backend service. The Design Service handles AI floor plan generation by calling Claude. The Construction Service manages site updates, photo uploads, and CV-based quality alerts. The Finance Service integrates with the escrow payment provider for milestone-based releases. The Schedule Service manages the Gantt chart, baseline timelines, and delay forecasts. Each service is independently deployable and can scale according to its specific load profile.

AI & Integration Layer

External AI and payment systems are integrated through dedicated adapter components. The Anthropic Claude API powers floor plan generation. A custom Computer Vision model hosted on AWS SageMaker analyses site photos for quality issues. RazorpayX Escrow handles all financial transactions. Firebase Cloud Messaging delivers push notifications across iOS, Android, and web.

Data Layer

PostgreSQL serves as the primary transactional database (using Amazon RDS with Multi-AZ failover for high availability). Amazon S3 stores all photos and generated floor plan files. Redis powers session

management, rate limiting, and cached reads of frequently accessed project data. All data is stored in the Mumbai AWS region for compliance with Indian data residency requirements.

7.2 Recommended Tech Stack

Table: Recommended Technology Stack

undefined	undefined	undefined
Customer Mobile	React Native + Expo	Cross-platform, fast iteration, OTA updates
PM Dashboard	Next.js 14 (App Router)	SEO-friendly, server components, Vercel hosting
Site Supervisor	React Native (Android only)	Camera & offline-first capabilities
API Gateway	Kong / AWS API Gateway	Centralised auth, rate limiting
Microservices	Node.js (TS) for most, Go for high-throughput	Match service to language strength
AI Layer	Anthropic Claude API	Best-in-class structured JSON generation
Computer Vision	PyTorch model on AWS SageMaker	Industry standard for production CV
Database	Postgres 16 (RDS Multi-AZ)	ACID transactions, JSON support for flexible schemas
Cache	Redis (ElastiCache)	Session, rate limits, hot reads
File Storage	AWS S3 (Mumbai)	Photo storage, durable, cheap
Queue	AWS SQS	Async CV processing, notification fan-out
Notifications	Firebase Cloud Messaging	Cross-platform push
Payments	RazorpayX Escrow	Indian-native, fastest integration
Maps	Mapbox / Google Maps	Site location, route planning
Analytics	PostHog (self-hosted)	Privacy-aware product analytics
Monitoring	Sentry + Datadog	Errors + APM
CI/CD	GitHub Actions + Argo CD	K8s-native deployments

8. Data Model

The core data model is organised around five primary entities. Each entity is described in plain language with its key attributes and relationships to other entities. All entities include standard audit fields (when created, when last updated, who created it, soft-delete flag) which are omitted from the descriptions below for brevity.

8.1 Project

The central entity representing a single home construction engagement. Every customer interaction, site update, payment, and milestone is linked back to a project.

Table: Project Entity — Core Fields

Field	Description
Project Name	Customer-visible name for the project (e.g. 'Sharma Residence')
Customer	Link to the homeowner who owns this project
Project Manager	Link to the assigned HomeBuild PM who oversees execution
Status	Current state: lead, in design, active, paused, completed, or cancelled
Plot Address	Full plot address as text
Plot Coordinates	Geo-location (latitude and longitude) for site navigation
Plot Length & Width	Plot dimensions in feet
Plot Facing	Cardinal direction the plot faces — North, South, East, or West
Total Floors	Number of floors planned (Ground only, G+1, G+2, or G+3)
BHK per Floor	Bedrooms per upper floor (1, 2, 3, or 4 BHK)
Selected Plan	Link to the chosen AI-generated floor plan
Budget Total	Agreed project budget in Indian Lakhs
Start Date	Project kickoff date
Expected Completion	Originally agreed completion date
Actual Completion	Date the project actually finished (filled in on handover)

8.2 Floor Plan

Stores each AI-generated floor plan option presented to a customer. Multiple plans are generated per project; one is eventually selected by the customer for construction.

Table: Floor Plan Entity — Core Fields

Field	Description
Project Link	Which project this plan belongs to
Variant	Plan identifier — A, B, or C (Symmetrical, Open, or Compact)
Name	Human-readable plan name
Vastu Score	1–10 rating of Vastu Shastra compliance
Efficiency Score	1–10 rating of usable space optimisation
Total Square Footage	Total built-up area across all floors
Cost per Square Foot	Construction rate based on budget tier
Total Cost	Estimated total cost in Indian Lakhs
Highlights	Three short descriptive bullets about the plan's strengths
Vastu Notes	Plain-text explanation of Vastu zones applied
Room Layout Data	Detailed room positions and dimensions for each floor
AI Generation Metadata	Which AI model and version produced this plan, for auditing
Is Selected	Whether the customer chose this plan for construction

8.3 Site Update

Each daily progress entry submitted by a site supervisor. Updates form the chronological site feed visible to both customer and project manager.

Table: Site Update Entity — Core Fields

Field	Description
Project Link	Which project this update belongs to
Supervisor	Site supervisor who submitted the update
Uploaded At	Timestamp of submission
Work Category	Type of work — e.g. 'Slab Casting', 'Brickwork', 'Plastering'
Location on Site	Where on the building the work was performed (e.g. 'Level 2, NE corner')
Notes	Supervisor's free-text notes about the day's progress
Photos	List of photo URLs attached to this update
Status	Work status — In Progress, Done, or Blocked
Milestone Reference	Optional link to the milestone this update contributes to

8.4 AI Alert

Quality issues detected automatically by the computer vision model when analysing uploaded site photos. Alerts drive the workflow for the project manager to investigate and resolve issues before they become rework.

Table: AI Alert Entity — Core Fields

Field	Description
Project Link	Which project this alert belongs to
Source Update Link	Which site update (and photo) triggered this alert
Category	Type of issue detected — e.g. 'Rebar Spacing', 'Concrete Curing'
Severity	Issue severity — Low, Medium, High, or Critical
Confidence Score	AI model's confidence in the detection, between 0 and 1
Description	Plain-language description of what was detected
Location in Photo	Coordinates on the photo highlighting the issue area
Status	Workflow state — New, Acknowledged, Assigned, Resolved, or False Positive
Assigned To	Who is responsible for resolving this alert
Resolved At	When the alert was marked resolved
Resolution Photo	Photo demonstrating the fix

8.5 Milestone & Payment

Project milestones that trigger escrow payment releases. Each milestone has a defined percentage of the total project cost, verification criteria, and payment workflow.

Table: Milestone Entity — Core Fields

Field	Description
Project Link	Which project this milestone belongs to
Sequence	Order in the milestone schedule (1, 2, 3, etc.)
Name	Milestone name (e.g. 'Foundation Complete')
Release Percentage	Percentage of total budget released on completion
Release Amount	Actual payment amount in Indian Lakhs
Due Date	Originally scheduled completion date
Status	Upcoming, Pending verification, or Paid
PM Verification Time	When the project manager signed off on completion
Customer Approval Time	When the customer approved the release
Payment Transaction ID	Reference to the escrow provider's transaction record

9. Third-Party Integrations

Table: Third-Party Integration Priorities

undefined	undefined	undefined
Anthropic Claude API	P0	Floor plan generation, alert text summarisation
AWS SageMaker	P0	Custom CV model hosting
RazorpayX Escrow	P0	Milestone-based payment escrow
Firebase Auth + FCM	P0	User authentication, push notifications
Google Maps / Mapbox	P1	Plot location, navigation for site supervisors
Twilio / MSG91	P1	OTP-based authentication, SMS alerts
DigiLocker	P1	Customer KYC document fetching
MagicBricks / 99acres	P2	Plot listings cross-reference for lead enrichment
Infra.Market / Moglix	P2	Bulk material procurement and pricing
Zoho Books / Tally	P2	Accounting integration for contractor invoicing

10. Phased Roadmap & Milestones

Phase 1 — Foundation (Months 1–3)

- Tech stack setup, CI/CD pipeline, monorepo structure
- Customer auth, project onboarding flow
- Design Module: AI floor plan generation (v1, no Vastu engine)
- Basic PM dashboard with project list
- Internal pilot with 5 projects (founder-led)

Phase 2 — Core MVP (Months 4–6)

- Construction Module: photo upload + site feed (no CV yet)
- Finance Module: budget breakdown + manual milestone tracking
- Schedule Module: Gantt chart with baseline + actual
- Site supervisor Android app (basic)
- Vastu compliance scoring layer for Design
- Onboard 20 paying customers in pilot city

Phase 3 — AI Layer (Months 7–9)

- Computer vision model training & deployment (3–4 alert categories)
- Forecast algorithm for Schedule Module
- RazorpayX Escrow integration (replace manual milestone tracking)
- Multi-language UI: Hindi, Tamil, Telugu
- Expand to second pilot city

Phase 4 — Scale (Months 10–12)

- CV model expansion to 8+ alert categories
- Material procurement marketplace integration
- AMC product launch for completed projects
- Open referral programme
- Series A fundraise preparation
- Reach ₹15 Cr ARR run rate

11. Success Metrics (KPIs)

Table: Year 1 Success Metrics and KPIs

undefined	undefined	undefined
Growth	Active projects	500+
Growth	Monthly new project signups	60+ by Month 12
Engagement	% of customers viewing site updates weekly	80%+
Engagement	Daily active site supervisors	150+
Quality	On-time milestone completion rate	85%+
Quality	Customer NPS	>60
AI Value	% projects using AI-generated plan	70%+
AI Value	CV alert true positive rate	>85%
Financial	Average margin per project	₹3.5L+
Financial	GMV (Gross Merchandise Value)	₹250 Cr+
Financial	Revenue (project fees + materials)	₹20 Cr+
Retention	% customers signing AMC post handover	40%+
Referral	% projects from existing customer referral	30%+

12. Risks & Mitigation

Table: Project Risks and Mitigation Strategies

undefined	undefined	undefined
AI floor plans rejected by traditional clients	High	Always pair AI output with optional architect review; offer Vastu compliance scoring as non-negotiable trust signal
Computer vision false positives erode PM trust	High	Confidence threshold 0.75+; PM feedback loop to retrain model weekly; transparent confidence scores shown
Contractor network quality inconsistency	High	Mandatory vetting process; 90-day probation with rating; tie milestone payments to PM quality sign-off
Regulatory delays (BBMP, BDA approvals)	Medium	Hire local liaison architects; build approval status into Schedule Module; set realistic baselines
Escrow payment provider downtime	Medium	Multi-provider strategy (Razorpay + Cashfree); fallback to manual reconciliation with reservation
Customer churn during long construction cycle	Medium	Weekly automated progress reports; AMC upsell as continuity hook
New entrants targeting the same segment	Medium	Lock in city-level operations partnerships early; build defensive contractor exclusivity; double down on Vastu and regional language advantages
Monsoon delays cascading across project pipeline	Low	Build 15% monsoon buffer into all baselines; document weather-linked clauses in customer agreement

13. Out of Scope (v1)

The following are explicitly excluded from v1 to ensure focus. They may be considered in v2:

- 3D model rendering of generated floor plans (only 2D in v1)
- Augmented reality (AR) walkthroughs of designs
- Direct contractor marketplace (we manage them, not customers)
- Real-time video streaming from sites (only photos)
- Drone-based site surveys
- Sub-contracting marketplace for specialty trades (paint, tiles, etc.)
- White-label / B2B SaaS offering to other construction firms
- Insurance product bundling

14. Open Questions

The following decisions require alignment before development starts:

- Final choice between Claude API vs hybrid LLM strategy (cost vs quality trade-off)
- Buy vs build computer vision model (Roboflow templates vs custom training)
- Pilot city selection: Coimbatore, Indore, or Surat — research recommends Coimbatore
- Contractor compensation structure: salaried PMs vs revenue-share
- Material procurement: own warehouse vs Infra.Market drop-ship partnership
- Customer agreement: standard contract or city-specific legal review
- Brand name and visual identity (currently working name: 'HomeBuild')

Next Steps

1. Engineering & Design review of this PRD within 5 working days. 2. Technical spike on AI floor plan generation accuracy (1 week). 3. Pilot city contractor partnership outreach (parallel track). 4. Final PRD sign-off and kickoff sprint planning.